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# Total (restrained) domination in unit disk graphs

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ABSTRACT

The minimum total domination problem and the minimum total restrained domination problem are classical combinatorial optimization problems. In this paper, we first show that the decision versions of both the minimum total domination problem and the minimum total restrained domination problem are NP-complete for grid graphs (a subclass of unit disk graphs), thereby strengthening the result on the total domination decision problem in unit disk graphs established by Jena et al. (2021). Then, we present a linear-time 6.387-approximation algorithm for the minimum total domination problem in unit disk graphs, improving upon the previous algorithm with a ratio of 8 by Jena et al. (2021). Finally, we propose a linear-time 10.387-approximation algorithm and a PTAS for the minimum total restrained domination problem in unit disk graphs.



## 1. Introduction

The exploration of domination in graph theory has garnered significant attention, with a wealth of relevant literature available in the two books by Haynes et al. [8,9]. Variations of domination have also been extensively studied [2,6,10,12].

Throughout this paper, let $G=(V,E)$ be a simple graph with no isolated vertices. Given a vertex $v \in V$, the *open neighborhood* $N_G(v)$ is defined as the set $\{u \in V | uv \in E\}$, and the degree of $v$, denoted by $|N_G(v)|$, is the cardinality of this set. Additionally, the *closed neighborhood* of $v$ is represented as $N_G[v]=N(v)\cup\{v\}$. We simplify $N_G(v)$ and $N_G[v]$ to $N(v)$ and $N[v]$, respectively. For two distinct vertices $u$ and $v$, the *distance* $d(u,v)$ between $u$ and $v$ is the length of a shortest path between $u$ and $v$. Let $N_2[v]$ denote the set of all vertices at distance at most 2 from $v$ in $G$. For a set $D \subseteq V$, the graph $G[D]$ induced by $D$ is the graph with vertex set $D$ and edge set $\{uv \in E | u, v \in D\}$. A *leaf* of $G$ is a vertex of degree 1, a *support vertex* is a vertex adjacent to a leaf.

A set $I \subseteq V$ in graph $G$ is considered an *independent set* (IS) if it contains no adjacent vertices. It is referred to as a *maximal independent set* (MIS) if, for any vertex $v \in V \setminus I$, $I \cup \{v\}$ is not an IS. The *independence number* $\alpha(G)$ is the maximum cardinality among all independent sets of $G$.

A set $D \subseteq V$ is considered a *dominating set* (resp. *total dominating set* (TDS), introduced by Cockayne et al. [4]) of $G$ if every vertex in $V \setminus D$ (resp. $V$) is adjacent to a vertex in $D$. Additionally, if every vertex in $V$ is adjacent to a vertex in $D$, and every vertex in $V \setminus D$ is adjacent to a vertex in $V \setminus D$, then $D$ is a *total restrained dominating set* (TRDS), a concept introduced by Ma et al. [12]. Clearly, $D$ is a TRDS if and only if $D$ is a dominating set and $G[D]$ and $G[V \setminus D]$ have no isolated vertices.

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We say that a vertex $u \in V$ is *dominated* by a vertex $v \in V$ if $uv \in E$. The minimum cardinality of total dominating sets (resp. total restrained dominating sets) of $G$ is called the *total domination number* (resp. *total restrained domination number*) of $G$, denoted by $\gamma_t(G)$ (resp. $\gamma_{tr}(G)$). By the definitions of TDS and TRDS, we easily obtain the following two observations:

**Observation 1.1.** For any graph $G$, $\gamma_t(G) \le \gamma_{tr}(G)$.

**Observation 1.2.** Every TRDS of a graph $G$ contains all leaves and support vertices in $G$.

The problem of finding a TDS (resp. a TRDS) of $G$ with the minimum cardinality is called the Minimum Total Domination (or, for short, MTD) problem (resp. the Minimum Total Restrained Domination (or, for short, MTRD) problem). Given a graph $G$ and a positive integer $k$, the Total Domination Decision (or, for short, TDD) problem (resp. the Total Restrained Domination Decision (or, for short, TRDD) problem) is to decide whether there exists a TDS of $G$ (resp. a TRDS) of cardinality at most $k$.

A unit disk is a disk with a diameter of 1. A *unit disk representation* of a graph $G = (V, E)$ is a set $S = \{S_u\}_{u\in V}$ of unit disks, such that $uv \in E$ if and only if $S_u \cap S_v \neq \emptyset$. A graph is a *unit disk graph* (or, for short, UDG) if it has a unit disk representation. Alternatively, a graph $G$ is a UDG if there is an edge between two vertices if and only if their Euclidean distance is at most 1. A grid graph is a UDG where all the disks in the disk representation have centers with integer coordinates. Equivalently, a grid graph can be defined as an induced subgraph of an $(h, w)$-grid for some positive integers $h$ and $w$, where the $(h, w)$-grid is an undirected graph $G$ with a vertex set $\{(x, y) : x, y \in \mathbb{Z}, 1 \le x \le h, 1 \le y \le w\}$ and an edge set $\{(u, v)(x, y) : |u - x| + |v - y| = 1\}$.

The TRDD problem is NP-complete for bipartite graphs and chordal graphs [12], planar graphs and split graphs [1], circle graphs, undirected path graphs, S-CB graphs and C-CB graphs [16]. Yang et al. [16] proved that the problem of deciding whether $\gamma_t(G) = \gamma_{tr}(G)$ is NP-hard even when $G$ is restricted to planar bipartite graphs with $\Delta(G) \le 5$. In [10] Henning summarized a large number of conclusions about the MTD problem. It is well known that the decision version of the minimum domination problem for a graph is NP-hard in UDGs [3]. In [11] Jena et al. showed that the TDD problem is NP-complete in UDGs and given a linear time 8-approximation algorithm for the MTD problem in UDGs. In this paper, we strengthen their results and prove the TDD problem is NP-complete even for grid graphs (a subclass of UDGs) and present a linear time 6.387-approximation algorithm for the MTD problem in UDGs.

The present paper is organized as follows. In Section 2, we prove that the TDD problem and the TRDD problem are NP-complete for grid graphs. In Section 3, we give a linear time 6.387-approximation algorithm for the MTD problem in UDGs. In Section 4, we propose a linear time 10.387-approximation algorithm and a PTAS for the MTRD problem in UDGs. Finally, in Section 5, we present some concluding remarks and future work.

## 2. NP-completeness

In this section, we show that the TRDD problem and the TDD problem are NP-complete in grid graphs. Here we use the following NP-complete problem, due to Dahlhaus et al. [5], for the reduction.

**Planar Exactly 3-Bounded 3-Sat (PE 3-B 3-S)**

**Instance:** A formula $\Phi$ with a variable set $X$ and a clause set $C$ such that, firstly, each variable appears in exactly 3 clauses, each clause has size at most 3, and secondly, the bipartite graph $B$ with bipartition $(X, C)$, obtained by connecting an edge between $x \in X$ and $c \in C$ if $c$ contains one of the literals $x$ or $\bar{x}$, is planar. Notice that the degree of each vertex in $B$ is at most 3.

**Question:** Is $\Phi$ satisfiable?

Next, we give a theorem by Valiant [14] which will be used in the following proof.

**Theorem 2.1.** *[14]. A planar graph $G$ with vertex set $V$ and maximum degree 4 can be embedded in the plane using $O(|V|^2)$ area in such a way that its vertices are located at points with integer coordinates and its edges are drawn as a sequence of connected line segments of the form $x = i$ (vertical) or $y = j$ (horizontal) for integers $i$ and $j$.*

Now, we are prepared to prove the main results in this section.

**Theorem 2.2.** *The* TRDD *problem is NP-complete for grid graphs.*

**Proof.** It is not hard to see that this decision problem is in NP. To complete the complexity proof, we use a reduction from the PE 3-B 3-S problem. Consider a set of clauses $C = \{c_1, c_2, \cdots, c_t\}$ with variables $X = \{x_1, x_2, \cdots, x_n\}$, for some positive integers $t$ and $n$, as an input for the PE 3-B 3-S problem. Now we construct a graph $G = (V, E)$, in the following way. For any variable $x_i \in X$, we introduce a vertex $p_i$, for any clause $c_j \in C$, we introduce a vertex $q_j$. If $x_i \in c_j$ or $\overline{x_i} \in c_j$, then we add an edge $p_iq_j$. (See Fig. 1 (a) for an example. Here $G$ amounts to $B$ in the definition of the PE 3-B 3-S problem.) Then $G$ is planar with $3n$ edges, is of maximum degree 3 and further, there is a kind of a planar embedding in Theorem 2.1, denoted by $G_1$ of $G$ (see Fig. 1 (b)). Now we construct a grid graph $G_2$ from $G_1$, using the following procedure (see Fig. 3 (b) for an example).

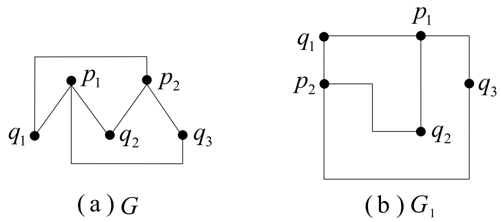


**Fig. 1.** (a) A graph $G$ constructed from a formula $\Phi = (x_1 \vee x_2) \wedge (x_1 \vee \overline{x_2}) \wedge (\overline{x_1} \vee x_2)$. (b) An embedding graph $G_1$ of $G$.

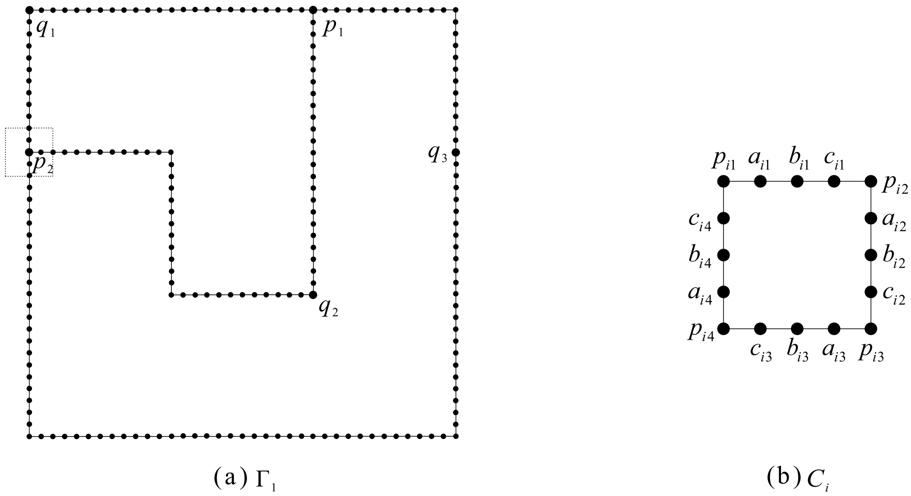


**Fig. 2.** (a) A drawing of $\Gamma_1$, the box of the dotted line is the graph $G[N_2[p_2]]$; (b) A $4 \times 4$ square $C_i$ corresponding to $p_i$.

**The construction of $G_2$:**

(1) Expand the embedding by 12 times, add a vertex at each grid point, the resulted embedding is denoted by $\Gamma_1$ (see Fig. 2 (a)), and for each vertex $p_i$ $(1 \le i \le n)$ (resp. $q_j$ $(1 \le j \le t)$), label still as $p_i$ (resp. $q_j$) the vertex at which $p_i$ (resp. $q_j$) is located in $\Gamma_1$. Each edge $q_jp_i$ in $G$ represents a path from $q_j$ to $p_i$ in $\Gamma_1$, and the path is denoted by $K_{ji}$. Obviously, the length of the $K_{ji}$ is divided by 12, we divide the path $K_{ji}$ into $k_{ji}$ subpaths with each being of length 12.

(2) Since the maximum degree of $G$ is 3, $q_j$ for $j \in \{1, \cdots, t\}$ has a blank space in $\Gamma_1$ in either the horizontal or vertical direction, on which we add a path of length 3 with an endpoint $q_j$ and label the vertices as $q_{j1}, q_{j2}, q_{j3}$ in turn (see Fig. 3 (a)). For each $1 \le i \le n$, replace the induced subgraph $G[N_2[p_i]]$ (see Fig. 2 (a)) with the $4 \times 4$ square $C_i$ with labels as described in Fig. 2 (b), such that the position of the center of the square $C_i$ is located at the position of $p_i$. Denote the resulted drawing by $\Gamma_2$ (see Fig. 3 (a)). For each edge $q_jp_i$ in $G$, let $K'_{ij}$ represent the path from $q_j$ to $b_{is}$ in $\Gamma_2$, where some integer $s \in \{1, 2, 3, 4\}$, $(V(K'_{ji}) = (V(K_{ji}) \setminus N_2[p_i]) \cup b_{is})$.

(3) Let $q_jp_i$ be an edge in $G$.

If $\overline{x_i} \in c_j$ and, then the subpath of length 10 containing $b_{i1}$ (resp. $b_{i3}$, $b_{i4}$, $b_{i2}$) on $K'_{ji}$ in $\Gamma_2$, then replace the corresponding path in $\Gamma_2$ with the shape as presented in Fig. 4 (a) (resp. (b), (c), (d)) such that the shape pastes with $C_i$ at the vertex $c_{i1}$ (resp. $c_{i3}, c_{i4}, c_{i2}$).

If $x_i \in c_j$ and, then the subpath of length 10 containing $b_{i1}$ (resp. $b_{i3}$, $b_{i4}$, $b_{i2}$) on the $K'_{ji}$ in $\Gamma_2$, is replaced by the shape as presented in Fig. 4 (e) (resp. (f), (g), (h)) such that the shape pastes with $C_i$ at the vertex $a_{i1}$ (resp. $a_{i3}, a_{i4}, a_{i2}$). Denote the resulted drawing by $G_2$, see Fig. 3 (b).

Clearly, $G_2$ is a grid graph and can be constructed in polynomial time. Next, we will show that $\Phi$ is satisfiable if and only if $\gamma_{tr}(G_2) \le 17n + 6k + 2t$, where $k = \sum_{q_jp_i \in E} k_{ji}$. For each $1 \le i \le n$, let $A_1^i = \{p_{i1}, p_{i2}, p_{i3}, p_{i4}, a_{i1}, a_{i2}, a_{i3}, a_{i4}\}$, $A_2^i = \{a_{i1}, a_{i2}, a_{i3}, a_{i4}, b_{i1}, b_{i2}, b_{i3}, b_{i4}\}$, $B_1^i = V(C_i) \setminus A_1^i$ and $B_2^i = V(C_i) \setminus A_2^i$. Each edge $q_jp_i$ in $G$ corresponds to a path $q_jd_1^{ji} \cdots d_{12k_{ji}}^{ji} a_{is}(c_{is})$, where some integer $s \in \{1, 2, 3, 4\}$, and two $P_2$-paths in $G_2$ (see Fig. 5 for an example). Let $L_{ji}$ be the set of the four vertices of the two $P_2$-paths, $P_{ji} = \{d_1^{ji}, d_2^{ji}, \cdots, d_{12k_{ji}}^{ji}\}$, $S_1^{ji} = \{d_{4s-3}^{ji}, d_{4s-2}^{ji} | s = 1, 2, \cdots, 3k_{ji} - 1\} \cup \{d_{12k_{ji}-3}^{ji}\}$ and $S_2^{ji} = \{d_{4s-2}^{ji}, d_{4s-1}^{ji} | s = 1, 2, \cdots, 3k_{ji} - 1\} \cup \{d_{12k_{ji}-1}^{ji}\}$.

Suppose now that $\Phi$ is satisfiable with an assignment. We build a TRDS $D$ of size $17n + 6k + 2t$ in $G_2$ as follows (see Fig. 3 (b) for an example). Firstly, add $\{q_{j3}, q_{j2}\}$ for each $1 \le j \le t$ and $L_{ji}$ for each $q_jp_i \in E$ into $D$. Secondly, for each $1 \le i \le n$, if $x_i$ is true, then add $A_1^i$ and $S_1^{ji}$ (resp. $A_1^i$ and $S_2^{ji}$) when $x_i \in c_j$ (resp. $\overline{x_i} \in c_j$) into $D$. If $x_i$ is false, then add $B_1^i$

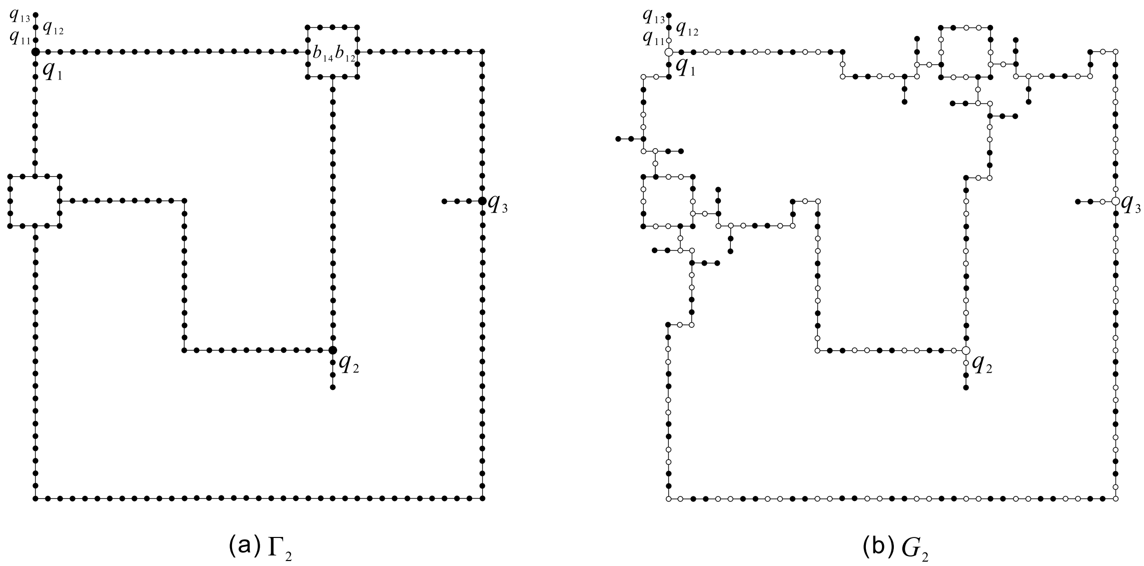


**Fig. 3.** (a) A drawing of $\Gamma_2$. (b) A grid graph $G_2$ obtained from $G_1$. The formula $\Phi = (x_1 \vee x_2) \wedge (x_1 \vee \overline{x_2}) \wedge (\overline{x_1} \vee x_2)$ is satisfiable with an assignment where $x_1$ is true and $x_2$ is true. The value of $k$ for $G_2$ is 17. The set of black vertices that we construct in $G_2$ forms a TRDS $D$ of $G_2$ with $|D| = 142$.

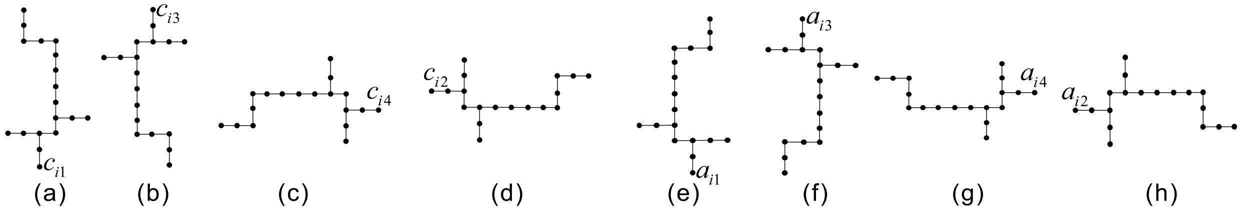


**Fig. 4.** The shapes (a)-(d) correspond to the case that $\overline{x_i} \in c_j$; the shapes (e)-(h) correspond to the case that $x_i \in c_j$.

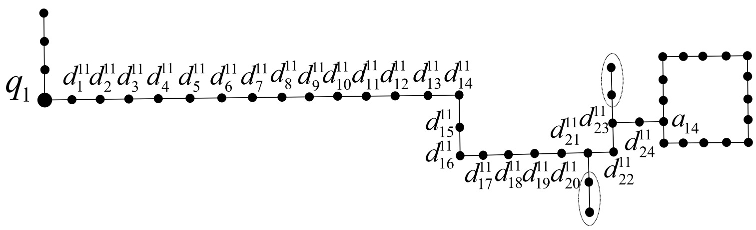


**Fig. 5.** An edge $q_1 p_1$ in $G$ corresponds to a path $q_1 d_1^{11} \cdots d_{24}^{11} a_{14}$, and the $P_2$-path is indicated by an ellipse.

and $S_2^{ji}$ (resp. $B_1^i$ and $S_1^{ji}$) when $x_i \in c_j$ (resp. $\overline{x_i} \in c_j$) into $D$. It can be seen from the above construction that, for each edge $q_j p_i$ in $G$, $6k_{ji} + 3$ vertices of $V(G_2)$ are put into $D$. For each vertex $p_i$ in $G$, add 8 vertices ($A_1^i$ or $B_1^i$) in $V(G_2)$ to $D$ and for each $q_j$ in $G$, add two vertices ($q_{j2}, q_{j3}$) in $V(G_2)$ to $D$. Thus

$$\begin{aligned} |D| &= \sum_{q_j p_i \in E} (6k_{ji} + 3) + 8n + 2t \\ &= 6k + 3 \cdot 3n + 8n + 2t \\ &= 17n + 6k + 2t. \end{aligned}$$

Next we show that $D$ is a TRDS of $G_2$. By the construction of $D$, for each vertex $v \in V(G_2) \setminus D$, $v$ is adjacent to a vertex in $V(G_2) \setminus D$. It is sufficient to show that for any $v \in V(G_2)$, $v$ is adjacent to a vertex in $D$. Obviously, each vertex $v \in V(G_2) \setminus \{q_j | 1 \leq j \leq t\}$ is adjacent to a vertex in $D$. Since $\Phi$ is satisfiable, each clause $c_j$ has a literal $x_i$ or $\overline{x_i}$ which evaluates true, thus $q_j$ is adjacent to the vertex $d_1^{ji}$ in $D$, where $d_1^{ji} \in P_{ji}$.

Conversely, let $T$ be a TRDS of size at most $17n + 6k + 2t$ in $G_2$, such that $|T \cap I|$ is as small as possible, where $I = \{d_{12k_{ji}}^{ji} | q_j p_i \in E\}$. By Observation 1.2, $(\cup_{q_j p_i \in E} L_{ji}) \cup \{q_{j2}, q_{j3} | 1 \leq j \leq t\} \subseteq T$.

**Claim 2.1.** $T \cap I = \emptyset$.

**Proof.** Suppose to the contrary that there exist $1 \le i \le n$ and $1 \le j \le t$ such that $d^{ji}_{12k_{ji}} \in T$. Without loss of generality, assume $d^{ji}_{12k_{ji}}$ is adjacent to $c_{i1}$; the other cases are similar. If $c_{i1} \notin T$, since $d^{ji}_{12k_{ji}} \in T$, $d^{ji}_{12k_{ji}-1} \in T$. If $b_{i1} \in T$ or $p_{i2} \in T$, let $T' = T \setminus \{d^{ji}_{12k_{ji}}\}$. Otherwise, if $b_{i1}, p_{i2} \notin T$, then $a_{i1}, a_{i2} \in T$. Let $T' = (T \setminus \{d^{ji}_{12k_{ji}}\}) \cup \{p_{i2}\}$. In all cases, $T'$ is a TRDS of $G_2$ and $|T'| \le |T|$, but $|T' \cap I| < |T \cap I|$, a contradiction. Thus, $c_{i1} \in T$. Considering the construction of $G_2$, both $p_{is}$ and $b_{is}$ have a degree of 2, and at most one of $a_{is}$ or $c_{is}$ has a degree of 3, where $s \in \{1, 2, 3, 4\}$. Since the degree of $c_{i1}$ is 3, the degree of $a_{i1}$ is 2.

Suppose $d^{ji}_{12k_{ji}-1} \notin T$. If $b_{i1} \in T$ or $p_{i2} \in T$, let $T' = T \setminus \{d^{ji}_{12k_{ji}}\}$. Otherwise, since $T$ is a TRDS and the degree of $a_{i1}$ is 2, $b_{i1}, p_{i2}, a_{i1}, a_{i2} \notin T$, $p_{i1}, c_{i4} \in T$. Thus, if $b_{i4} \in T$, then let $T' = (T \setminus \{p_{i1}, d^{ji}_{12k_{ji}}\}) \cup \{b_{i1}\}$. If $b_{i4} \notin T$, then $a_{i4} \notin T$. Next, let's first assume that $p_{i4} \in T$, then $c_{i3} \in T$. If $b_{i3} \in T$, then let $T' = (T \setminus \{p_{i1}, p_{i4}, d^{ji}_{12k_{ji}}\}) \cup \{b_{i1}, b_{i4}\}$. If $b_{i3} \notin T$, then $a_{i3} \notin T$. If $p_{i3} \notin T$, then let $T' = (T \setminus \{p_{i1}, p_{i4}, d^{ji}_{12k_{ji}}\}) \cup \{b_{i1}, b_{i3}, b_{i4}\}$. Otherwise $p_{i3} \in T$, $c_{i2} \in T$. If $b_{i2} \in T$, then let $T' = (T \setminus \{p_{i1}, p_{i3}, p_{i4}, d^{ji}_{12k_{ji}}\}) \cup \{b_{i1}, b_{i3}, b_{i4}\}$. If $b_{i2} \notin T$, then let $T' = (T \setminus \{d^{ji}_{12k_{ji}}\}) \cup \{p_{i2}\}$. Now, suppose $p_{i4} \notin T$, then let $T' = (T \setminus \{p_{i1}, d^{ji}_{12k_{ji}}\}) \cup \{b_{i1}, b_{i4}\}$. In all cases, $T'$ is a TRDS of $G_2$ and $|T'| \le |T|$, but $|T' \cap I| < |T \cap I|$, a contradiction. Thus, $d^{ji}_{12k_{ji}-1} \in T$.

Next, we first claim that $d^{ji}_{12k_{ji}-3} \notin T$. If $d^{ji}_{12k_{ji}-3} \in T$, since $d^{ji}_{12k_{ji}-1} \in T$, it follows that $d^{ji}_{12k_{ji}-2} \in T$. Therefore, if $b_{i1}$ or $p_{i2} \in T$, we let $T' = T \setminus \{d^{ji}_{12k_{ji}-2}, d^{ji}_{12k_{ji}-1}, d^{ji}_{12k_{ji}}\}$. Otherwise $p_{i2}, b_{i1} \notin T$, then $a_{i1} \notin T$. Since the degree of $a_{i1}$ is 2, $p_{i1} \in T$, let $T' = T \setminus \{d^{ji}_{12k_{ji}-2}, d^{ji}_{12k_{ji}-1}, d^{ji}_{12k_{ji}}\} \cup \{a_{i1}, b_{i1}\}$. Obviously, since $L_{ji} \in T$, in both cases $T'$ is a TRDS of $G_2$ and $|T'| < |T|$, but $|T' \cap I| < |T \cap I|$, which leads to a contradiction. Thus, we conclude that $d^{ji}_{12k_{ji}-3} \notin T$.

Now, we assume that $d^{ji}_{12k_{ji}-4} \notin T$. The case of $d^{ji}_{12k_{ji}-4} \in T$ is similar to the case when $d^{ji}_{12k_{ji}-4} \notin T$. If $b_{i1}$ or $p_{i2} \in T$, then let $T' = (T \setminus \{d^{ji}_{12k_{ji}-1}, d^{ji}_{12k_{ji}}\}) \cup \{d^{ji}_{12k_{ji}-3}, d^{ji}_{12k_{ji}-4}\}$. Otherwise, since $T$ is a TRDS and the degree of $a_{i1}$ is 2, $b_{i1}, p_{i2}, a_{i1}, a_{i2} \notin T$, we have $p_{i1}, c_{i4} \in T$. Thus if $b_{i4} \in T$, then let $T' = (T \setminus \{p_{i1}, d^{ji}_{12k_{ji}}, d^{ji}_{12k_{ji}-1}\}) \cup \{b_{i1}, d^{ji}_{12k_{ji}-4}, d^{ji}_{12k_{ji}-3}\}$. If $b_{i4} \notin T$, then $a_{i4} \notin T$. Let's first consider the case where $p_{i4} \notin T$. Since $b_{i4}, a_{i4} \notin T$ and in order to dominate $a_{i4}$, there is a vertex $d^{ti}_{12k_{ti}} \in N(a_{i4}) \setminus \{b_{i4}, p_{i4}\}$ (there is an edge $q_t p_i$ in $G$ corresponding to a path $q_1 d^{ti}_1 \cdots d^{ti}_{12k_{ti}} a_{i4}$) such that $d^{ti}_{12k_{ti}} \in T$. Since $a_{i4} \notin T$, $d^{ti}_{12k_{ti}-1} \in T$. Let $T' = (T \setminus \{p_{i1}, d^{ti}_{12k_{ti}}, d^{ji}_{12k_{ji}}, d^{ji}_{12k_{ji}-1}\}) \cup \{b_{i1}, b_{i4}, d^{ji}_{12k_{ji}-4}, d^{ji}_{12k_{ji}-3}\}$.

Next, we consider $p_{i4} \in T$. Since $p_{i4} \in T$, $c_{i3} \in T$. If $b_{i3} \in T$, then let $T' = (T \setminus \{p_{i1}, p_{i4}, d^{ji}_{12k_{ji}-1}, d^{ji}_{12k_{ji}}\}) \cup \{b_{i1}, b_{i4}, d^{ji}_{12k_{ji}-4}, d^{ji}_{12k_{ji}-3}\}$. Now, if $b_{i3} \notin T$, then $a_{i3} \notin T$. Firstly, if $p_{i3} \in T$, then $c_{i2} \in T$. If $b_{i2} \in T$, then let $T' = (T \setminus \{b_{i2}, d^{ji}_{12k_{ji}}, d^{ji}_{12k_{ji}-1}\}) \cup \{p_{i2}, d^{ji}_{12k_{ji}-4}, d^{ji}_{12k_{ji}-3}\}$. Otherwise since $p_{i2}, a_{i2}, b_{i2} \notin T$, there is $d^{ti}_{12k_{ti}} \in N(a_{i2}) \setminus \{b_{i2}, p_{i2}\}$ such that $d^{ti}_{12k_{ti}} \in T$, then $d^{ti}_{12k_{ti}-1} \in T$. Let $T' = (T \setminus \{d^{ti}_{12k_{ti}}, d^{ji}_{12k_{ji}}, d^{ji}_{12k_{ji}-1}\}) \cup \{p_{i2}, d^{ji}_{12k_{ji}-4}, d^{ji}_{12k_{ji}-3}\}$. Finally, if $p_{i3} \notin T$, since $b_{i3}, a_{i3} \notin T$, there is $d^{ti}_{12k_{ti}} \in N(a_{i3}) \setminus \{b_{i3}, p_{i3}\}$ such that $d^{ti}_{12k_{ti}} \in T$, then $d^{ti}_{12k_{ti}-1} \in T$. Let $T' = (T \setminus \{p_{i1}, p_{i4}, d^{ti}_{12k_{ti}}, d^{ji}_{12k_{ji}}, d^{ji}_{12k_{ji}-1}\}) \cup \{b_{i1}, b_{i3}, b_{i4}, d^{ji}_{12k_{ji}-4}, d^{ji}_{12k_{ji}-3}\}$. In all cases $T'$ is a TRDS of $G_2$ and $|T'| \le |T|$, but $|T' \cap I| < |T \cap I|$, a contradiction. Therefore, $T \cap I = \emptyset$. □

Since $T \cap I = \emptyset$ by Claim 2.1 and $C_i$ is a cycle with 16 vertices, $|V(C_i) \cap T| \ge 8$ for each $1 \le i \le n$. We consider the path $q_j d^{ji}_1 \cdots d^{ji}_{12k_{ji}} a_{is}(c_{is})$, where some integer $s \in \{1, 2, 3, 4\}$, and the corresponding two $P_2$-path, in $G_2$, in order to dominate $d^{ji}_{12k_{ji}-2}$, then $d^{ji}_{12k_{ji}-1} \in T$ or $d^{ji}_{12k_{ji}-3} \in T$. Assume that $d^{ji}_{12k_{ji}-3} \in T$ ($d^{ji}_{12k_{ji}-1} \in T$ is similar), since $d^{ji}_1 \cdots d^{ji}_{12k_{ji}-4}$ is a path with $12k_{ji} - 4$ vertices, $|T \cap \{d^{ji}_1, \cdots, d^{ji}_{12k_{ji}-4}\}| \ge 6k_{ji} - 2$. Then $|T \cap P_{ji}| \ge 6k_{ji} - 1$ by $d^{ji}_{12k_{ji}-3} \in T$. Note that $((\cup_{q_j p_i \in E} L_{ji}) \cup \{q_{j2}, q_{j3} | 1 \le j \le t\}) \subseteq T$. Since $G$ has $3n$ edges, $|(\cup_{q_j p_i \in E} L_{ji}) \cup \{q_{j2}, q_{j3} | 1 \le j \le t\}| = 4 \times 3n + 2t$. Then $|T| \ge 4 \times 3n + 2t + 8n + \sum_{q_j p_i \in E}(6k_{ji} - 1) = 17n + 6k + 2t$. Since $|T| = 17n + 6k + 2t$, it follows that $|T \cap P_{ji}| = 6k_{ji} - 1$ for each $q_j p_i \in E$, $|V(C_i) \cap T| = 8$ for each $1 \le i \le n$ and $q_j, q_{j1} \notin T$ for each $1 \le j \le t$. From the above discussion, we get the following observation.

**Observation 2.1.** For each $1 \le i \le n$, $T$ contains exactly one of the sets $A^i_1, A^i_2, B^i_1, B^i_2$.

**Claim 2.2.** *For each path $q_j d^{ji}_1 \cdots d^{ji}_{12k_{ji}} c_{is}(a_{is})$ in $G_2$, if $d^{ji}_{12k_{ji}}$ is adjacent to $c_{is}$ (resp. $a_{is}$) and $d^{ji}_1 \in T$, then $B^i_1$ or $B^i_2$ is contained in $T$ (resp. $A^i_1$ or $A^i_2$ in $T$), where $s = 1, 2, 3, 4$.*

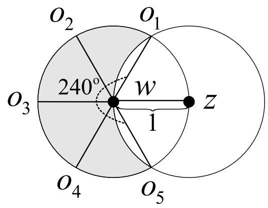


**Fig. 6.** The shaded area indicates the area $D(w) \setminus D(z)$.

**Proof.** Since $d_1^{ji} \in T$ and $q_j \notin T$, $d_2^{ji} \in T$. Then $T \cap P_{ji} = S_1^{ji}$ by $|T \cap P_{ji}| = 6k_{ji} - 1$. Thus $d_{12k_{ji}-1}^{ji} \notin T$. To dominate $d_{12k_{ji}}^{ji}$, $c_{is}$ must be included in $T$ (resp. $a_{is}$ must be included in $T$), it follows that $B_1^i$ or $B_2^i$ (resp. $A_1^i$ or $A_2^i$) is contained in $T$ by Observation 2.1. □

In summary, we can define an assignment of $\Phi$ as follows: set $x_i$ to be true if $A_1^i$ or $A_2^i$ is contained in $T$, to be false otherwise. Since for each $1 \le j \le t$, $q_j, q_{j1} \notin T$, there is $1 \le i_0 \le n$ such that $q_j$ is dominated by $d_1^{ji_0}$ in $T$, where $d_1^{ji_0} \in P_{ji_0}$. If $x_{i_0} \in c_j$, it means that $d_{12k_{ji_0}}^{ji_0}$ is adjacent to $a_{i_0 s}$, where $s \in \{1,2,3,4\}$, then $A_1^i$ or $A_2^i$ corresponding to $p_{i_0}$ in $T$ by Claim 2.2, it follows that $x_{i_0}$ is true. If $\overline{x_{i_0}} \in c_j$, it means that $d_{12k_{ji_0}}^{ji_0}$ is adjacent to $c_{i_0 s}$, where $s \in \{1,2,3,4\}$, then $B_1^i$ or $B_2^i$ corresponding to $p_{i_0}$ is contained in $T$ by Claim 2.2, it follows that $\overline{x_{i_0}}$ is true. Consequently, $\Phi$ satisfies all clauses. □

Using a similar proof as above, where we modify the construction of $G_2$, in the following way we change the $P_3$-path added to $q_j$ into $P_4$-path for each $1 \le j \le t$ and for each $p_j p_i \in E$, change the $P_2$-path added to $d_{12k_{ji}-1}^{ji}$ and $d_{12k_{ji}-3}^{ji}$ into $P_3$-path. We obtain the following theorem.

**Theorem 2.3.** *The* TDD *problem is NP-complete for grid graphs.*

## 3. A 6.387-approximation algorithm of MTD in UDGs

In this section, we design an approximation algorithm for the MTD problem in UDGs. The approximate ratio of our proposed algorithm is 6.387 which improves the 10-factor approximation algorithm given in [13] and a 8-factor approximation algorithm given in [11]. We call a disk with radius 1 centered at $x$ the *adjacent area* of $x$, denoted by $D(x)$. Generally, for a graph $G$, its *adjacent area* is the union of all adjacent areas of its vertices. For UDGs, there are two lemmas.

**Lemma 3.1.** *[15] There are at most five independent vertices in the adjacent area of a vertex.*

**Lemma 3.2.** *[15] There are at most eight independent vertices in the adjacent area of two adjacent vertices.*

**Theorem 3.1.** *The size of any IS in a UDG $G$ is at most $4\gamma_t(G)$.*

**Proof.** Let $I$ be any IS of $G$ and $D^*$ be any minimum TDS of $G$. Let $G_1, \cdots, G_q$ be all components of $G[D^*]$ such that $|V(G_i)| = 2$ for any $1 \le i \le p$ (such components may not exist) and $|V(G_i)| \ge 3$ for any $p+1 \le i \le q$, where $p, q$ are two integers. If $p = q$, then the proof is completed by Lemma 3.2. Next, we assume that $p \ne q$. We first prove the following claim which will help us to the later proof.

**Claim 3.1.** *For any two vertices $w, z \in D^*$ with $wz \in E$, the area $D(w) \setminus D(z)$ contains at most 4 independent vertices.*

**Proof.** Since $wz \in E$, the distance between $w$ and $z$ is at most 1. When the distance between $w$ and $z$ is 1, the area $D(w) \setminus D(z)$ is the largest, see Fig. 6. Let $o_1$, $o_5$ be the intersection points of the two disks $D(w)$ and $D(z)$. Then $\angle o_1 w o_5 = 240°$. Thus the area $D(w) \setminus D(z)$ can be divided into 4 smaller area $o_1wo_2, o_2wo_3, o_3wo_4, o_4wo_5$ such that $\angle o_1wo_2 = \angle o_2wo_3 = \angle o_3wo_4 = \angle o_4wo_5 = 60°$, see Fig. 6. The distance of any two vertices in same area $o_1wo_2$ or $o_2wo_3$ or $o_3wo_4$ or $o_4wo_5$ is at most 1, so there are at most 4 independent vertices in the area $D(w) \setminus D(z)$. □

Let $T_i$ be a spanning tree of $G_i$ for any $1 \le i \le q$ and $T$ be a forest with vertex set $V(T) = \cup_{1 \le i \le q} V(T_i)$ and edge set $E(T) = \cup_{1 \le i \le q} E(T_i)$. Since all vertices of $G$ lie in the adjacent area of $T$ by the definitions of TDS and UDG, we only need to show that there are at most $4|D^*| = 4|V(T)|$ independent vertices in the adjacent area of $T$. We will show by induction on $|V(T)|$. Since $p \ne q$, $|V(T)| \ge 3$. If $|V(T)| = 3$, then $q = 1$ by the definition of TDS, it follows that $T = T_1$. Let $V(T_1) = \{x, y, z\}$ and $E(T_1) = \{xy, yz\}$. The adjacent area $D(x) \cup D(y) \cup D(z)$ of $T_1$ can be divided into two disjoint areas

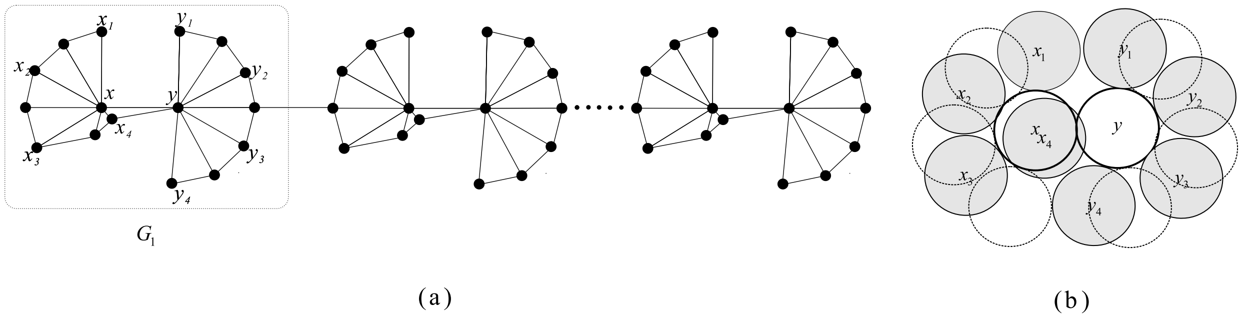


**Fig. 7.** (a) A class of UDGs $G$ with $\alpha(G) = 4\gamma_t(G) = 4\gamma_{tr}(G)$, $G$ is constructed from a list copies of $G_1$ by connecting a pair of consecutive copies with an edge as showed. (b) A disk representation of $G_1$. The eight shadowed disks and two bold disks are corresponding to a maximum IS and a minimum TDS (TRDS) in $G_1$.

$D(x) \cup D(y)$ and $D(z) \setminus D(y)$ by $E(T_1) = \{xy, yz\}$. By Lemma 3.2 (resp. Claim 3.1), there are at most 8 (resp. 4) independent vertices in $D(x) \cup D(y)$ (resp. $D(z) \setminus D(y)$). Thus the adjacent area of $T_1$ contains at most 12 $(= 4|D^*|)$ independent vertices.

Next, we assume $|V(T)| \geq 4$. In fact, since $p \neq q$, $|V(T)| \geq 5$ and $|V(T_q)| \geq 3$. Thus there is a non-leaf vertex $v$ in $T_q$ adjacent to at least one leaf. Let $u$ be a leaf neighbor of $v$ in $T_q$. By Claim 3.1, $D(u) \setminus D(v)$ contains at most 4 independent vertices. By the induction hypothesis, there are at most $4(|V(T)| - 1)$ independent vertices in the adjacent area of $T - \{u\}$. Therefore, the adjacent area of $T$ contains at most $4(|V(T)| - 1) + 4 = 4|V(T)|$ independent vertices. □

By combined with Observation 1.1, we have:

**Corollary 3.1.** *The size of any IS in a UDG G is at most* $4\gamma_{tr}(G)$.

The upper bounds in Theorem 3.1 and Corollary 3.1 are tight (see Fig. 7 for an example).

Given a connected UDG $G = (V, E)$. Let $I$ be a MIS of $G$. For a set $S \subseteq V$, define $f(S)$ be the number of isolated vertices in $G[I \cup S]$. Define $\Delta_X f(S) = f(S) - f(S \cup X)$ for any subset $X$ of $V$. If $X = \{x\}$, then simply write $\Delta_x f(S)$ instead of $\Delta_{\{x\}} f(S)$. In the following we give a greedy algorithm for the MTD problem in UDGs (i.e., Algorithm 1).

**Algorithm 1 APPROX-TOTAL-DOM-SET(G).**

**Input:** A connected UDG $G = (V, E)$, a MIS $I$ of $G$.
**Output:** A TDS $D$ of $G$.

```
    Set D ← ∅, S ← ∅;
while f(S) > 0 do
    select a vertex x ∈ V \ I that maximizes Δ_x f(S);
    S ← S ∪ {x};
end while
D ← S ∪ I;
return D.
```

Next we will show that the time complexity of Algorithm 1 is $O(n+m)$, where $n = |V|$ and $m = |E|$.

**Theorem 3.2.** *Algorithm 1 runs in time* $O(n+m)$, *where* $n = |V|$ *and* $m = |E|$.

**Proof.** The time of Algorithm 1 is mainly spent in the process of finding $S$. For each vertex $x \in V \setminus I$, compute $N_I(x)$, where $N_I(x)$ is a set of the neighbors of $x$ in $I$. For each vertex $u \in I$, compute $N_{V\setminus I}(u)$, where $N_{V\setminus I}(u)$ is a set of the neighbors of $u$ in $V \setminus I$. By adjacency list this can be implemented in $O(n+m)$ time. Let $M_0 = \emptyset$ and $M_t = \{x \in V \setminus I || N_I(x)| = t\}$. Since $I$ is a MIS of $G$, $1 \leq t$. By Lemma 3.1, $t \leq 5$. We use Algorithm 2 to compute the set $S$.

Suppose the while loop iterates $g$ times, obviously $g \leq n$. Note that $1 \leq m_i \leq 5$ for each $1 \leq i \leq g$. In the first iteration of Algorithm 2, each vertex in $M_{m_1}$ connects the most number of isolated vertices in $G[I]$. We consider any $i$th iteration, where $2 \leq i \leq g$. We have chosen $i-1$ vertices $x_1, \cdots, x_{i-1}$ before the $i$th iteration. Let $S_{i-1} = \{x_1, \cdots, x_{i-1}\}$. By the process of the algorithm, each vertex in $M_{m_i}$ connects the most number of isolated vertices in $G[I \cup S_{i-1}]$. Thus in each while loop, we select a vertex $x_i$ in $M_{m_i}$ that maximize $\Delta_x f(S_{i-1})$.

In Algorithm 2, through the analysis of the first paragraph, line 2 takes $O(n+m)$ time. The number of iterations of the line 3 loop is $g$. It's easy to know that line 4 and line 13 take constant time. In line 5, since $|N_I(x_i)| \leq 5$, $N_I(x_i) \setminus I'$ can be determined in constant time. For any vertex $x$ in $I$, $N_x = N_{V\setminus I}(x)$ can be determined in $O(n+m)$ time. Therefore, when loop $g$ times lines 4-5 and line 13 take $O(n+m)$ time in Algorithm 2.

Now let's consider lines 6-12 in the line 3 loop. In the $i$th iteration, where $1 \leq i \leq g$, let $n_i = \sum_{1 \leq j \leq m_i} |N_{u_j}|$. Since lines 8-10 spend constant time, it only needs to consider how many times lines 8-10 occurred, in other words, the time of lines

**Algorithm 2 COMPUTE-S.**

**Input:** A connected UDG $G=(V,E)$, a MIS $I$ of $G$.
**Output:** A set $S$.
1: Initialize $S=\emptyset$, $I'=\emptyset$, $i=1$;
2: Compute $N_I(x)$ for all $x\in V\setminus I$ and $N_{V\setminus I}(u)$ for all $u\in I$. Let $M_0=\emptyset$, and compute $M_t=\{x\in V\setminus I||N_I(x)|=t\}$, where $1\le t\le 5$;
3: **while** $f(S)>0$ **do**
4: $\quad m_i=\max_{1\le t\le 5}\{t|M_t\neq\emptyset\}$, choose a vertex $x_i$ from $M_{m_i}$;
5: $\quad N_I(x_i)\setminus I'=\{u_1,\cdots,u_{m_i}\}$, $N_{u_j}=N_{V\setminus I}(u_j)$ for all $1\le j\le m_i$;
6: $\quad$ **for** $1\le j\le m_i$ **do**
7: $\quad\quad$ **for** each $x\in N_{u_j}$ **do**
8: $\quad\quad\quad$ there is $1\le t\le m_i$ such that $x\in M_t$;
9: $\quad\quad\quad$ $M_t\leftarrow M_t\setminus\{x\}$;
10: $\quad\quad\quad$ $M_{t-1}\leftarrow M_{t-1}\cup\{x\}$;
11: $\quad\quad$ **end for**
12: $\quad$ **end for**
13: $\quad S=S\cup\{x_i\}$, $I'=I'\cup N_I(x_i)$, $i=i+1$;
14: **end while**
15: **return** $S$.

6-12 depends on $n_i$. Thus the total time of lines 6-12 in line 3 loop depends on $\sum_{i=1}^{g} n_i$. Within Algorithm 2, each while loop selects a vertex from $V\setminus I$, and the iterations conclude when $f(S)=0$. Additionally, since $N_u=N_{V\setminus I}(u)$, $u\in N_I(x_i)\setminus I'$, for each $u\in I$, $N_u$ is considered exactly once. Consequently, $\sum_{i=1}^{g} n_i$ represents the total number of edges from vertices in $I$ to vertices in $V\setminus I$. Therefore, $\sum_{i=1}^{g} n_i=\sum_{v\in I}|N_{V\setminus I}(v)|=\sum_{u\in V\setminus I}|N_I(u)|$. Since each vertex in $V\setminus I$ has at most 5 neighbors in $I$ by Lemma 3.1, $\sum_{i=1}^{g} n_i\le 5|V\setminus I|\le 5n$. It's means that the total time of lines 6-12 in line 3 loop is $O(n)$. Thus Algorithm 2 runs in time $O(n+m)$. That is Algorithm 1 runs in time $O(n+m)$. □

In the rest of the section, we analyze the correctness and approximation ratio of Algorithm 1.

**Theorem 3.3.** *The output $D$ of Algorithm 1 is a TDS of $G$ and the approximation ratio is 6.387.*

**Proof.** Since every vertex in $V\setminus I$ has a neighbor in $I$, and every vertex in $I$ has a neighbor in $S$, $D$ is a TDS of $G$. Next we will show that the approximation ratio is 6.387. Let $I$ be the MIS of $G$ of the input in Algorithm 1. Assume $x_1,x_2,\cdots,x_g$ are vertices in $S=D\setminus I$, in the order of their selection into the set. Let $S^*=\{y_1,y_2,\cdots,y_m\}$ be a minimum TDS for $G$. Denote $S_i=\{x_1,x_2,...,x_i\}$ and $S_j^*=\{y_1,y_2,\cdots,y_j\}$, where $1\le i\le g$ and $1\le j\le m$. Denote $S_0=S_0^*=\emptyset$.

Since $I$ is a MIS of $G$, any isolated vertex in $G[I\cup B]$ only appears in $I$ for any set $B\subseteq V$. Thus if a vertex is isolated in one of $G[I\cup S_{i-1}\cup S_{j-1}^*]$ or $G[I\cup S_{i-1}\cup\{y_j\}]$, then it is also isolated in $G[I\cup S_{i-1}]$. If a vertex is isolated in both $G[I\cup S_{i-1}\cup S_{j-1}^*]$ and $G[I\cup S_{i-1}\cup\{y_j\}]$, then it is also isolated in $G[I\cup S_{i-1}]$ and $G[I\cup S_{i-1}\cup S_{j-1}^*\cup\{y_j\}]$. Thus for any $1\le j\le m$,

$$f(S_{i-1}\cup S_{j-1}^*)+f(S_{i-1}\cup\{y_j\})\le f(S_{i-1})+f(S_{i-1}\cup S_{j-1}^*\cup\{y_j\}),$$

that is,

$$\Delta_{y_j}f(S_{i-1}\cup S_{j-1}^*)\le\Delta_{y_j}f(S_{i-1}). \tag{1}$$

Note that for any $1\le i\le g$, we have,

$$f(S_{i-1})-f(S_{i-1}\cup S^*)=f(S_{i-1})-f(S_{i-1}\cup\{y_1\})+f(S_{i-1}\cup\{y_1\})-f(S_{i-1}\cup\{y_1,y_2\})+f(S_{i-1}\cup\{y_1,y_2\})-f(S_{i-1}\cup\{y_1,y_2,y_3\})+\cdots+f(S_{i-1}\cup\{y_1,y_2,\cdots,y_{m-1}\})-f(S_{i-1}\cup S^*\}).$$

Thus,

$$\Delta_{S^*}f(S_{i-1})=\sum_{j=1}^{m}\Delta_{y_j}f(S_{i-1}\cup S_{j-1}^*). \tag{2}$$

Since $S^*$ is a TDS for $G$, there are no isolated vertices in $G[I\cup S^*]$, thus $f(S_{i-1}\cup S^*)=0$. Combining (1) and (2), we get for any $1\le i\le g$,

$$\begin{aligned}f(S_{i-1})-0&=f(S_{i-1})-f(S_{i-1}\cup S^*)\\&=\Delta_{S^*}f(S_{i-1})\\&=\sum_{j=1}^{m}\Delta_{y_j}f(S_{i-1}\cup S_{j-1}^*)\end{aligned}$$

$$\leq \sum_{j=1}^{m} \Delta_{y_j} f(S_{i-1})$$

Then there exists a vertex $y_j \in S^*$ such that

$$\Delta_{y_j} f(S_{i-1}) \geq \frac{f(S_{i-1})}{m}.$$

By the greedy strategy of Algorithm 1,

$$\Delta_{x_i} f(S_{i-1}) \geq \Delta_{y_j} f(S_{i-1}) \geq \frac{f(S_{i-1})}{m}.$$

It implies that,

$$f(S_i) \leq f(S_{i-1}) - \frac{f(S_{i-1})}{m} = f(S_{i-1})(1 - \frac{1}{m}). \tag{3}$$

If $g \leq tm$, where $t > 0$, then $|D| = |S| + |I| \leq tm + 4m = (t+4)m$ by Theorem 3.1, thus the proof is already done. So we assume that $g > tm$.

Solving the recurrence relation (3), we have

$$f(S_i) \leq f(S_0)(1 - \frac{1}{m})^i. \tag{4}$$

When $g - tm$ vertices are added to $S$, there are still isolated vertices in $G[I \cup S_{g-tm}]$, so $f(S_{g-tm}) \geq g - (g - tm) = tm$. Set $i = g - tm$. Then $tm \leq f(S_i) \leq |I|(1 - \frac{1}{m})^i$ by inequality (4) and $f(S_0) = |I|$. Since $(1 - \frac{1}{m})^i \leq e^{-i/m}$, we obtain $i \leq m \cdot \ln \frac{|I|}{tm}$. It follows that $g = tm + i \leq m(t + \ln \frac{|I|}{tm}) \leq m(t + \ln(\frac{4}{t}))$ by Theorem 3.1, then $|D| = |S| + |I| \leq (t + \ln(\frac{4}{t}))m + 4m$. It's easy to see that $t + \ln\left(\frac{4}{t}\right)$ attains its minimum value at $t = 1$, which is 2.387. Let $t = 1$, combining the above statements: $|D| \leq 6.387m$. □

## 4. Approximation algorithms of MTRD in UDGs

In Section 2 we proved that the TRDD problem is NP-complete for grid graphs. In this section, we present a simple linear time 10.387-approximation algorithm (Algorithm 3) and a PTAS (Algorithm 4) for the MTRD problem in UDGs. We first show the following lemma:

**Lemma 4.1.** *For any graph $G = (V, E)$, let $D'$ be a TDS of $G$ and $I'$ be the set of vertices which are isolated in $G[V \setminus D']$. Then $I'$ is an IS of $G$ and $D = D' \cup I'$ is a TRDS of $G$.*

**Proof.** If there are two vertices $u, v \in I'$ such that $uv \in E$, then $u, v \in V \setminus D'$ are not isolated in $G[V \setminus D']$, a contradiction. Thus $I'$ is an IS of $G$. Now we show that $D$ is a TRDS of $G$. Since $D'$ is a TDS of $G$, it is sufficient to prove that each vertex $u \in V \setminus D$ has a neighbor in $V \setminus D$. Since $u \in V \setminus D$, $u \in V \setminus D'$ and $u \notin I'$. By the definition of $I'$ and $u \notin I'$, $u$ has a neighbor $v \in V \setminus D'$. Then $v$ is not isolated in $G[V \setminus D']$. Thus $v \notin I'$, it follows that $v \in V \setminus D$. □

Now we present the linear time 10.387-approximation algorithm for the MTRD problem in UDGs.

**Algorithm 3 APPROX-TOTAL-RESTRAINED-DOM-SET(G).**

**Input:** A connected UDG $G = (V, E)$.
**Output:** A TRDS $D$ of $G$.
1: A TDS $D'$ obtained from Algorithm 1.
2: $I' \leftarrow \emptyset$.
3: **for** each isolated vertex $w$ in $G[V \setminus D']$ **do**
4: $\quad I' \leftarrow I' \cup \{w\}$;
5: **end for**
6: **return** $D = D' \cup I'$

**Theorem 4.1.** *Algorithm 3 is a 10.387-approximation for the MTRD problem in UDGs which runs in time $O(n+m)$, where $n = |V|$ and $m = |E|$.*

**Proof.** By Lemma 4.1, $D = D' \cup I'$ is a TRDS of $G$. By Observation 1.1 and Theorem 3.3, $|D'| \leq 6.387\gamma_t(G) \leq 6.387\gamma_{tr}(G)$. By Lemma 4.1, $I'$ is an IS of $G$. Then by Corollary 3.1, $|I'| \leq 4\gamma_{tr}(G)$. Thus $|D| = |D'| + |I'| \leq 10.387\gamma_{tr}(G)$. By Theorem 3.2, the time to compute $D'$ is $O(n+m)$. We can determine whether a vertex in $V \setminus D'$ is isolated in $G[V \setminus D']$ according to the adjacency list of $G[V \setminus D']$. Thus the lines 3-5 take $O(n+m)$ time. Then Algorithm 3 runs in time $O(n+m)$. □

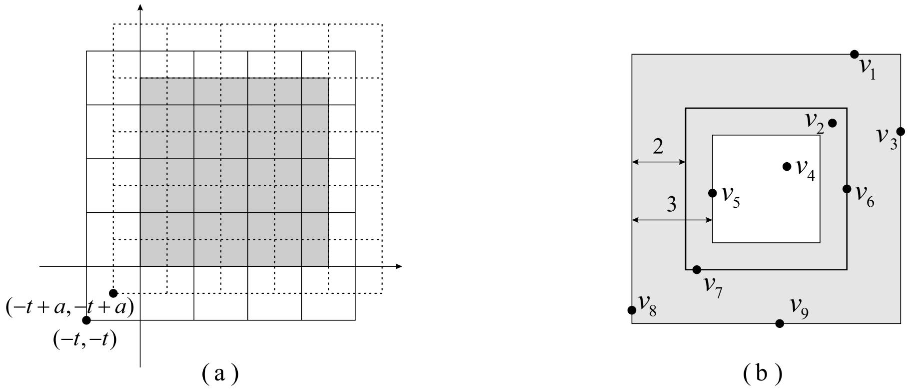


**Fig. 8.** (a) The shaded area (resp. solid grid, dashed grid) indicates the square $Q$ (resp. partition $P(0)$, partition $P(a)$). (b) The outermost (resp. middle) square indicates a cell $e$ (the square $e^c$). $B_e$ is the set of vertices in (b) excluding the vertices in the white area, as well as the vertices on the top and right boundaries of the outermost square and the boundary of the innermost square.

In the following we will show that exists a PTAS for the MTRD problem in UDGs.

For a connected UDG $G = (V, E)$, let $Q$ be a minimal rectangle that contains all vertices in $V$. Assume $Q = \{(x, y)|0 \leq x \leq q, 0 \leq y \leq q\}$, where $q$ is a positive integer. For a positive real number $\varepsilon < 1$, let $t = \lceil 747.864/\varepsilon \rceil$ and $p = \lfloor q/t \rfloor + 1$. Firstly, we enlarge the area of the square $Q$ to $Q' = \{(x, y)| -t \leq x \leq pt, -t \leq y \leq pt\}$. Secondly, we divide $Q'$ into cells of size $t \times t$. Define this partition as $P(0)$. For any $0 \leq a \leq t-1$, denote by $P(a)$ the partition obtained from $P(0)$ by shifting the left-bottom corner of $P(0)$ from $(-t, -t)$ to $(-t+a, -t+a)$, see Fig. 8 (a).

For each cell $e$ of a partition $P(a)$, exclude the top and right boundaries of $e$. Let $V(e)$ be the set of vertices contained in $e$ and $B_e$ (resp. $C_e$) be the *boundary set* (resp. *central set*) of $e$ which is a set of vertices contained in $e$ that have distance less than 3 (resp. at least 2) from the boundary of $e$. See Fig. 8 (b) for an example, where $\{v_2, v_4, \cdots, v_9\} \subseteq V(e)$, $\{v_2, v_6, v_7, v_8, v_9\} \subseteq B_e$ and $\{v_2, v_4, v_5, v_6, v_7\} \subseteq C_e$. The *boundary set* of a partition $P(a)$ is the union of the boundary sets of all cells in $P(a)$. A set $D_e \subseteq V(e)$ is called a *TRDS of* $C_e$ if each vertex in $C_e \cup D_e$ is adjacent to a vertex in $D_e$ and each vertex in $C_e \setminus D_e$ is adjacent to a vertex in $V(e) \setminus D_e$. Note that the definition of TRDS given in Section 1 is about a graph, the definition given here is about a set and the vertex in $D_e$ is allowed not to be in $C_e$.

**Algorithm 4 PTAS-TOTAL-RESTRAINED-DOM-SET(G).**

**Input:** A connected UDG $G = (V, E)$ with all vertices lying in $Q$ and a positive real number $\varepsilon < 1$.
**Output:** A TRDS $D$ of $G$.
1: Let $t = \lceil 747.864/\varepsilon \rceil$.
2: Use Algorithm 3 to compute a TRDS $D_0$ of $G$.
3: For each $a \in \{0, 1, \cdots, t-1\}$, denote by $D_0(a)$ the set of vertices in $D_0$ containing in the boundary set of $P(a)$.
Let $a' = \text{argmin}_{a \in \{0,1,\cdots,t-1\}} |D_0(a)|$.
4: For each cell $e$ of $P(a')$, compute a minimum TRDS $D_e$ of $C_e$.
5: Let $D' = D_0(a') \cup (\bigcup_{e \in P(a')} D_e)$.
6: Let $T$ be the set of vertices which are isolated in $G[V \setminus D']$.
7: Let $D = D' \cup T$.
**return** $D$

By Algorithm 4, we get the following theorem:

**Theorem 4.2.** *The output $D$ of Algorithm 4 is a TRDS of $G$. Moreover, each vertex in $T$ has a neighbor $D_0(a')$.*

**Proof.** By Lemma 4.1, we only need to show that $D'$ is a TDS of $G$. For any $v \in V$, there is a unique cell $e$ of the partition $P(a')$ such that $v \in V(e)$. First consider the case $v \in C_e$. By line 4, $D_e$ is a TRDS of $C_e$, then there is a neighbor of $v$ in $D_e$.

Now we consider the case $v \in B_e \setminus C_e$. Since $D_0$ is a TRDS of $G$, $v$ is dominated by a vertex $u \in D_0$. If $u \notin V(e)$, then there is a cell $e'$ of the partition $P(a')$ adjacent with $e$ such that $u \in V(e')$. Since the distance between $u$ and $v$ is at most 1, $u$ has distance less than 1 from the boundary of $e'$, it follows that $u \in B_{e'}$, then $u \in D_0 \cap B_{e'}$. The case that remains to be seen $u \in V(e)$. Since each vertex in $B_e \setminus C_e$ has distance less than 2 from the boundary of $e$ and the distance between $u$ and $v$ is at most 1, $u$ has distance less than 3 from the boundary of $e$, it follows that $u \in B_e$, then $u \in D_0 \cap B_e$. In either case, $u \in D_0(a')$. Thus $D$ is a TRDS of $G$.

For each vertex in $T$, if $v \in T \cap C_e$, then $v \notin D_e$. Then $v$ has a neighbor $u \in V(e) \setminus D_e$. Since $v \in T$, there is no neighbor of $v$ in $V \setminus D'$, then $u \in D'$. By $u \in V(e) \setminus D_e$, $u \in D_0(a')$. If $v \in T \cap (B_e \setminus C_e)$, then $v$ has distance at most 2 from the boundary of $e$, and since $D_0(a')$ the set of vertices in $D_0$ containing in the boundary set of $P(a')$, $v$ has a neighbor $u \in D_0(a')$. Thus each vertex in $T$ has a neighbor $D_0(a')$. $\square$

The following lemma is used to analyze the time complexity and approximation ratio of Algorithm 4.

**Lemma 4.2.** *Let $G=(V,E)$ be a UDG, $J$ a subset of $V$ and $I \subseteq V \setminus J$ an IS of $G$. If each vertex in $I$ has a neighbor in $J$, then $|I| \le 5|J|$*

**Proof.** Let $m$ be the number of edges between $I$ and $J$. Each vertex in $I$ has a neighbor in $J$, so $m \ge |I|$. Since $I$ is an IS of $G$, each vertex in $J$ has at most 5 neighbors in $I$ by Lemma 3.1, then $m \le 5|J|$. Thus $|I| \le 5|J|$. □

The next theorem shows that the time complexity of Algorithm 4 is $n^{O(\frac{1}{\varepsilon^2})}$.

**Theorem 4.3.** *The running time of Algorithm 4 is $n^{O(\frac{1}{\varepsilon^2})}$, where $n=|V|$.*

**Proof.** Lines 2 and 6 take $O(n+m)$ time by Theorem 4.1. Clearly, the most time-consuming part is the line 4. For each cell $e$ of the partition $P(a')$, shrinking $e$ by 2 units will obtain a square $e^c$ same as the center of $e$. Then $e^c$ has size $(t-4)\times(t-4)$ and all vertices in $C_e$ are contained in $e^c$, see Fig. 8 (b). Partition $e^c$ into smaller-cells of size $\frac{\sqrt{2}}{2}\times\frac{\sqrt{2}}{2}$ (the distance between any two vertices in same smaller-cell is at most 1). There are $\lceil\sqrt{2}(t-4)\rceil^2$ smaller-cells in $e^c$. For each non-empty smaller-cell, choose any two vertices in the smaller-cell if it contains at least 2 vertices, otherwise choose the only one vertex $v$ and any one vertex $v'$ in $N(v)$ (note that $v'$ exists by the connectedness of $G$ and $v' \in V(e)$). Let $J_e$ be the set of all chosen vertices. Since we choose exactly two vertices in any smaller-cell or choose the only one vertex $v$ and any one vertex $v'$ in $N(v)$ and there are $\lceil\sqrt{2}(t-4)\rceil^2$ smaller-cells in $e^c$, $|J_e| \le 2\lceil\sqrt{2}(t-4)\rceil^2$.

Since the distance between any two vertices in same smaller-cell is at most 1, any two vertices in a smaller-cell are adjacent. Then each vertex in $C_e \cup J_e$ is adjacent to a vertex in $J_e$. Let $I_e=\{v \in C_e \setminus J_e | v$ have no neighbor in $V(e)\setminus J_e\}$ and $J'_e = J_e \cup I_e$. Similar as Lemma 4.1, $I_e$ is an IS of $G[V(e)]$ and $J'_e$ is a TRDS of $C_e$. Since each vertex in $I_e$ has a neighbor in $J_e$, $|I_e| \le 5|J_e|$ by Lemma 4.2. Then $|J'_e| = |J_e| + |I_e| \le 6|J_e| \le 12\lceil\sqrt{2}(t-4)\rceil^2 = O(t^2)$. Thus a minimum TRDS $D_e$ of $C_e$ can be obtained by checking all subset in $V(e)$ with size at most $O(t^2)$, which takes $n_e^{O(t^2)}$ time, where $n_e=|V(e)|$. It follows that the total time of Algorithm 4 is at most $\sum_{e\in p(a')} n_e^{O(t^2)} = n^{O(t^2)} = n^{O(\frac{1}{\varepsilon^2})}$. □

Lemma 4.3 plays a key role in analyzing approximate ratio. And the proof of the Lemma 4.3 has been given in Du et al. (2012).

**Lemma 4.3.** *[7] Let $S \subseteq V$ and $S(a)$ be the set of vertices in $S$ containing in the boundary set of a partition $P(a)$, where $0 \le a \le t-1$. Then each vertex in $S$ belongs to at most 12 of the sets $S(0), S(1), \cdots, S(t-1)$.*

Finally, we analyze the approximate ratio of Algorithm 4.

**Theorem 4.4.** *Algorithm 4 is a $(1+\varepsilon)$-approximation for the MTRD problem in UDGs.*

**Proof.** Let $D^*$ be a minimum TRDS of $G$. Note that when $a$ runs over $0,1,\cdots,t-1$, by Lemma 4.3, each vertex in $D_0$ belongs to at most 12 of the sets $D_0(0), D_0(1), \cdots, D_0(t-1)$. Hence

$$|D_0(0)| + |D_0(1)| + \cdots + |D_0(t-1)| \le 12|D_0|,$$

and thus by the choice of $a'$ and Theorem 4.1,

$$|D_0(a')| \le \frac{12}{t}|D_0| \le \frac{12\times 10.387}{t}|D^*| \le \frac{\varepsilon}{6}|D^*|.$$

Consider the partition $P(a')$. For any cell $e$ of $P(a')$, let $I_e^* \subseteq D^* \cap V(e)$ be the set of vertices which have no neighbor in $D^* \cap V(e)$.

**Claim 4.1.** *$D_e^* = (D^* \cap V(e)) \setminus I_e^*$ is a TRDS of $C_e$.*

**Proof.** We first prove that for any $v \in C_e \cup D_e^*$, there is a neighbor $u \in D_e^*$ of $v$. For any $v \in D_e^*$, $v$ has a neighbor $u \in D^* \cap V(e)$ by $v \in D^*$ and $v \notin I_e^*$, thus $u \in D_e^*$. For any $v \in C_e \setminus D_e^*$, since $D^*$ is a TRDS of $G$, there is a neighbor $u \in D^*$ of $v$, further $u$ has a neighbor $w \in D^*$. Since $v \in C_e$, $u \in V(e)$ has distance at least 1 from the boundary of $e$. Then $w \in V(e)$, it follows that $u \notin I_e^*$. Thus $u \in D_e^*$. Next we prove that for any $v \in C_e \setminus D_e^*$ has a neighbor in $V(e)\setminus D_e^*$. For any $v \in C_e \setminus D_e^*$, we claim $v \notin D^*$. Otherwise $v \in D^*$, then $v \in I_e^*$. Since $v \in C_e$, by the above discussion, $v$ has a neighbor $u \in D_e^*$, a contradiction with $v \in I_e^*$. Since $D^*$ is a TRDS of $G$ and $v \notin D^*$, there is a neighbor $u \in V \setminus D^*$ of $v$. Since $v \in C_e$ and $u \notin D^*$, $u \in V(e)$ and $u \notin D_e^*$. Then $u \in V(e)\setminus D_e^*$. In summary, $D_e^*$ is a TRDS of $C_e$. □

By Claim 4.1 and $D_e$ is a minimum TRDS of $C_e$, $|D_e| \le |D_e^*| \le |D^* \cap V(e)|$. Then

$$|\bigcup_{e\in P(a')} D_e| = \sum_{e\in P(a')} |D_e| \le \sum_{e\in P(a')} |D^* \cap V(e)| = |\bigcup_{e\in P(a')} (D^* \cap V(e))| = |D^*|.$$

By Lemma 4.1, $T$ is an IS of $G$. By Theorem 4.2, each vertex in $T$ has a neighbor $D_0(a')$. Then by Lemma 4.2, $|T| \le 5|D_0(a')|$. Hence by the above inequalities

$$|D| \le |D_0(a')| + |\bigcup_{e\in P(a')} D_e| + |T| \le |D^*| + 6|D_0(a')| \le (1+\varepsilon)|D^*|. \quad \square$$

## 5. Conclusion

In this paper, we have proven the NP-completeness of the TDD problem and TRDD problem for grid graphs. Furthermore, we have presented algorithms for the MTD problem and MTRD problem. It is worth noting that the cartesian product of two paths, $P_m \times P_n$, is a subclass of the grid graph discussed in this paper. Therefore, it would be interesting to explore whether these two problems are also NP-complete for traditional grid graphs ($P_m \times P_n$).

## CRediT authorship contribution statement

**Yu Yang:** Writing – review & editing, Writing – original draft, Data curation. **Cai-Xia Wang:** Writing – review & editing, Writing – original draft, Data curation. **Shou-Jun Xu:** Writing – review & editing, Writing – original draft, Data curation.

## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

## Data availability

No data was used for the research described in the article.

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